ENGINEERING DRAWING WITH AUTOCAD

UNIT 2: PROJECTION OF POINTS AND LINES

Curriculum: Lovely Professional University | School of Computer Science & Engineering
Reference: Engineering Drawing with an Introduction to AutoCAD by Dhananjay Jolhe

>> 2.1 Principles of Quadrants and Orthographic Projections

Orthographic projection is a parallel projection technique where projection rays are perpendicular to the projection plane:
* Principal Planes of Projection: Horizontal Plane (HP) and Vertical Plane (VP). The intersection line of HP and VP is called the Reference Axis or Ground Line (XY line).
* Four Spatial Quadrants:
- 1st Quadrant: Above HP, In front of VP (Observer -> Object -> Plane). Standard in Indian / European engineering.
- 2nd Quadrant: Above HP, Behind VP (HP and VP overlap when rotated 90 deg clockwise; both views fall above XY).
- 3rd Quadrant: Below HP, Behind VP (Observer -> Plane -> Object). Standard in American / Canadian engineering.
- 4th Quadrant: Below HP, In front of VP (Both views fall below XY line when HP is rotated 90 deg clockwise).
* Standard Notation Rules:
- Point in space: Capital letter (e.g., A, B, P).
- Elevation / Front View (on VP): Lowercase with prime (e.g., a'', b'', p'').
- Plan / Top View (on HP): Lowercase without prime (e.g., a, b, p).
- Side / End View (on Profile Plane PP): Lowercase with double prime (e.g., a''', b'''').

>> 2.2 Projection of Points in Different Quadrants

* Case 1: Point in 1st Quadrant (d1 above HP, d2 in front of VP): a'' is d1 above XY; a is d2 below XY.
* Case 2: Point in 2nd Quadrant (d1 above HP, d2 behind VP): Both a'' (d1) and a (d2) are above XY.
* Case 3: Point in 3rd Quadrant (d1 below HP, d2 behind VP): a is d2 above XY; a'' is d1 below XY.
* Case 4: Point in 4th Quadrant (d1 below HP, d2 in front of VP): Both a'' (d1) and a

(d2) are below XY.
* Points in Principal Planes: If on HP, front view a'' lies directly on XY. If on VP, top view a lies directly on XY.

>> 2.3 Projection of Straight Lines: Geometric Orientations

A straight line is the shortest locus between two points in 3D space:
* 1. Line Parallel to both HP and VP: Both front and top views show True Length (TL) and are parallel to XY.
* 2. Line Perpendicular to HP: Top view is a point; Front view is a vertical line perpendicular to XY showing True Length.
* 3. Line Perpendicular to VP: Front view is a point; Top view is a vertical line perpendicular to XY showing True Length.
* 4. Line Inclined to HP and Parallel to VP: Front view shows True Length and true inclination angle theta. Top view is shorter than TL (apparent length) and parallel to XY.
* 5. Line Inclined to VP and Parallel to HP: Top view shows True Length and true inclination angle phi. Front view is apparent length and parallel to XY.
* 6. Line Inclined to both HP and VP (Oblique Line): Neither view shows True Length; both views are apparent lengths (Elevation Length EL and Plan Length PL) inclined at apparent angles alpha and beta.

>> 2.4 True Length and Traces of a Line

* Traces of a Straight Line:
- Horizontal Trace (HT): The point of intersection of the line (or line produced) with the Horizontal Plane. Since HT lies on HP, its front view h'' lies on the XY line.
- Vertical Trace (VT): The point of intersection of the line (or line produced) with the Vertical Plane. Since VT lies on VP, its top view v lies on the XY line.
* Methods to Determine True Length and Inclinations from Projections:
- 1. Rotating Line Method: Rotate elevation or plan view until parallel to XY line, then project to find True Length.
- 2. Trapezoid Method: Construct right-angled trapezoid using difference in distances of endpoints from reference plane.
- 3. Auxiliary Plane Method: Set up an auxiliary plane parallel to the apparent view.

>> 2.5 AutoCAD 2D Drawing Commands and Precision Drafting

* LINE (L): Creates straight line segments between coordinate points.

* CIRCLE (C): Options: Center-Radius, Center-Diameter, 2-Point, 3-Point, Tan-Tan-Radius (TTR).
* ARC (A): 3-point, Start-Center-End, Start-Center-Angle, Start-End-Radius.
* PLINE (PL): Polyline creating single contiguous 2D entity containing line and arc segments with varied widths.
* RECTANG (REC): Defined by two diagonal corner coordinates; supports Chamfer, Fillet, and Elevation.
* POLYGON (POL): Creates regular multisided polygons (3 to 1024 sides) Inscribed in circle or Circumscribed about circle.
* ELLIPSE (EL): Generates true mathematical ellipses by defining major axis endpoints and half of minor axis length.
* DIMSTYLE (D): Configures dimension styles, text fonts, arrow sizes, precision, and tolerance displays.

[PROBLEM 1: Multi-Quadrant Point Projection Analysis]

Problem Statement: Draw the projections of the following four points on a common reference XY line, taking a 30 mm spacing between projectors:
* Point A: 35 mm above HP and 25 mm in front of VP.
* Point B: 40 mm above HP and 30 mm behind VP.
* Point C: 25 mm below HP and 45 mm behind VP.
* Point D: 30 mm below HP and 20 mm in front of VP.

Step-by-Step Construction Procedure:
* 1. Draw horizontal reference line XY.
* 2. Projector for Point A (1st Quadrant): Draw vertical projector line. Mark a'' 35 mm above XY. Mark a 25 mm below XY.
* 3. Projector for Point B (2nd Quadrant, 30 mm right): Draw vertical projector. Mark b'' 40 mm above XY. Mark b 30 mm above XY (both above XY).
* 4. Projector for Point C (3rd Quadrant, 30 mm right): Draw vertical projector. Mark c 45 mm above XY (top view). Mark c'' 25 mm below XY (front view).
* 5. Projector for Point D (4th Quadrant, 30 mm right): Draw vertical projector. Mark d'' 30 mm below XY. Mark d 20 mm below XY (both below XY).
* 6. Verify with clear dimension callouts and perpendicular construction guidelines.

[PROBLEM 2: Oblique Straight Line by Rotating Line Method]

Problem Statement: A straight line AB has its end A 15 mm above HP and 20 mm in front of VP. The line is 75 mm long (True Length). It is inclined at 30 degrees to HP and 45

degrees to VP. Draw its front view and top view, determine apparent lengths and apparent angles, and locate its traces HT and VT.

Step-by-Step Solution & Rotating Line Protocol:

* Step 1: Establish Initial Position of Point A:
- Draw XY line. Mark a'' 15 mm above XY and a 20 mm below XY on the same vertical projector.
* Step 2: Locus Lines of End B:
- Assume line parallel to VP, inclined at theta = 30 deg to HP: Draw a''b1'' = 75 mm at 30 deg to horizontal. Height of b1'' gives vertical locus of B: h_B = 15 + 75 x sin(30) = 15 + 37.5 = 52.5 mm above XY.
- Assume line parallel to HP, inclined at phi = 45 deg to VP: Draw ab2 = 75 mm at 45 deg to horizontal. Distance of b2 gives horizontal locus of B: d_B = 20 + 75 x sin(45) = 20 + 53.03 = 73.03 mm below XY.
* Step 3: Project Apparent Lengths:
- Project b1'' down to horizontal projector through a to get plan length: PL = ab1 = 75 x cos(30) = 64.95 mm.
- Project b2 up to horizontal projector through a'' to get elevation length: EL = a''b2'' = 75 x cos(45) = 53.03 mm.
* Step 4: Final Views by Rotation:
- Center at a, radius ab1 = 64.95 mm, rotate arc to intersect locus of B (at 73.03 mm below XY) -> Locate final top view b. Connect a-b.
- Center at a'', radius a''b2'' = 53.03 mm, rotate arc to intersect locus of B'' (at 52.5 mm above XY) -> Locate final front view b''. Connect a''-b''.
* Step 5: Verification & Angles:
- Check that a'' and a lie on the same vertical projector, and b'' and b lie on the same vertical projector.
- Apparent angles measured: alpha approx 45 deg, beta approx 55 deg.
* Step 6: Traces HT and VT:
- Extend front view b''a'' to intersect XY at h''. Draw perpendicular projector from h'' to intersect extended top view ba at HT.
- Extend top view ba to intersect XY at v. Draw perpendicular projector from v to intersect extended front view b''a'' at VT.

[PROBLEM 3: Finding True Length and Inclinations Given Projections]

Problem Statement: The top view of a 75 mm long line AB measures 55 mm, while the length of its front view is 65 mm. End A is in the HP and 15 mm in front of VP. Determine the true inclination of the line with HP and VP.

Mathematical Derivation:

* 1. Given: TL = 75 mm, PL = 55 mm, EL = 65 mm. End A: a'' on XY (distance 0 from

HP), a is 15 mm below XY.
* 2. Calculate true angle with HP (theta):
- cos(theta) = PL / TL = 55 / 75 = 0.7333 -> theta = arccos(0.7333) = 42.83 degrees.
* 3. Calculate true angle with VP (phi):
- cos(phi) = EL / TL = 65 / 75 = 0.8667 -> phi = arccos(0.8667) = 29.93 degrees.
* 4. Graphical Construction:
- Draw XY line. Mark a'' on XY, a 15 mm below XY.
- Draw a''b1'' = 75 mm at 42.83 deg to XY. Draw ab2 = 75 mm at 29.93 deg to XY.
- Draw locus lines through b1'' and b2, then swing arcs to locate final points b'' and b.

[PROBLEM 4: AutoCAD 2D Drafting Command Sequence for Geometric Plate]

Problem Context: Draft a precision regular pentagon (side 40 mm) with an inscribed circle and circumscribed ellipse, annotated with standard dimension styles.
AutoCAD Command Execution Log:
* Step 1: POLYGON -> Enter number of sides: 5 -> Edge option (E) -> Specify first endpoint: 100,100 -> Specify second endpoint: @40<0.
* Step 2: CIRCLE -> 3P option -> Select three vertices of the pentagon to draw circumscribed circle; or 2P/Tangent to create inscribed circle.
* Step 3: ELLIPSE -> Axis endpoint 1: 50,100 -> Axis endpoint 2: 190,100 -> Other axis distance: 35.
* Step 4: DIMSTYLE -> Modify -> Symbols and Arrows -> Arrow size: 3.5 -> Text height: 3.5 -> Fit: Always keep text between ext lines.
* Step 5: DIMALIGNED -> Select pentagon edge -> Place dimension 10 mm away showing 40.00 mm.
* Step 6: DIMDIAMETER -> Select inscribed circle -> Callout generated: dia 55.06 mm.